6. Subspaces

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Lecture due Jan 24, 2024 20:30 IST

Introduction to vector spaces and subspaces

9:50 / 9:50 2.0x

alone, satisfies the rules.

Why's that?

Oh, it's too dumb to tell you.

If I took that and added it to itself, I'm still there.

If I took that and multiplied by 17, I'm still there.

So I've done the operations-- adding and multiplying

by numbers--

that are required, and I didn't go outside this one point

space.

So that's the littlest subspace.

The largest space is the whole thing.

And in between come whatever's in between.

OK.

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Subspaces are subsets of vector spaces that are by themselves vector spaces.

Example 6.1 Subspaces of $\mathbb{R}^2$ (it turns out that this is the complete list):

- $\{\mathbf{0}\}$ (the set containing only the origin),
- any line through the origin,
- the whole plane $\mathbb{R}^2$.

Example 6.2 Subspaces of $\mathbb{R}^3$ (again, the complete list):

- $\{\mathbf{0}\}$,
- any line through the origin,
- any plane through the origin,
- the whole space $\mathbb{R}^3$.

Example problem

1/1 point (graded)

The set of linear combinations of the vectors $\begin{pmatrix}1\\3\\0\end{pmatrix}$ and $\begin{pmatrix}1\\4\\5\end{pmatrix}$ is a subspace of $\mathbb{R}^3$.

0. The zero vector is obtained by taking the zero combination:

$$0\begin{pmatrix}1\\3\\ \cdots\end{pmatrix}+0\begin{pmatrix}1\\4\\ \cdots\end{pmatrix}=\begin{pmatrix}0\\0\\ \cdots\end{pmatrix}$$

Calculator

$$0\begin{pmatrix}3\\0\end{pmatrix}+0\begin{pmatrix}4\\5\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$$

1. If a vector $\mathbf{v}$ can be written as a linear combination of the two vectors:

$$\mathbf{v}=a\begin{pmatrix}1\\3\\0\end{pmatrix}+b\begin{pmatrix}1\\4\\5\end{pmatrix},$$

then so can $c\mathbf{v}$:

$$c\mathbf{v}=ca\begin{pmatrix}1\\3\\0\end{pmatrix}+cb\begin{pmatrix}1\\4\\5\end{pmatrix}.$$

2. If two vectors $\mathbf{v}$ and $\mathbf{w}$ can be written as a linear combination of the two vectors:

$$\mathbf{v}=a_1\begin{pmatrix}1\\3\\0\end{pmatrix}+b_1\begin{pmatrix}1\\4\\5\end{pmatrix},\qquad \mathbf{w}=a_2\begin{pmatrix}1\\3\\0\end{pmatrix}+b_2\begin{pmatrix}1\\4\\5\end{pmatrix}$$

then so can $\mathbf{v}+\mathbf{w}$:

$$\mathbf{v}+\mathbf{w}=(a_1+a_2)\begin{pmatrix}1\\3\\0\end{pmatrix}+(b_1+b_2)\begin{pmatrix}1\\4\\5\end{pmatrix}.$$

This vector space is what kind of subspace of $\mathbb{R}^3$?

- ( ) The point $\{\mathbf{0}\}$
- ( ) A line through the origin.
- (•) A plane through the origin.
- ( ) The whole space $\mathbb{R}^3$.

✔

Solution:

The linear combinations of the two vectors $\begin{pmatrix}1\\3\\0\end{pmatrix}$ and $\begin{pmatrix}1\\4\\5\end{pmatrix}$ forms a plane in $\mathbb{R}^3$, because these two vectors do not lie on the same line through the origin.

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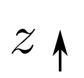
Answers are displayed within the problem

Subspaces concept check

1/1 point (graded)

Is the plane below a subspace of $\mathbb{R}^3$?



Calculator

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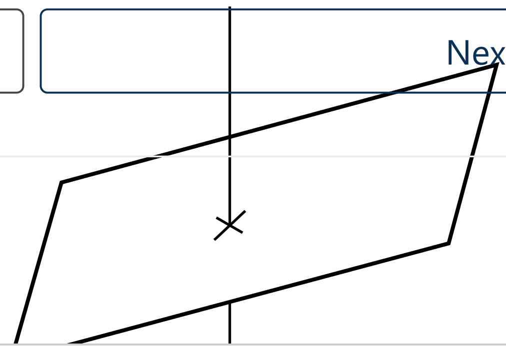

Yes.

No.

Cannot be determined without more information.

Solution:

The plane in the image above is not a subspace of $\mathbb{R}^3$ because it is not passing through the origin.

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